

CAP213:PRINCIPLES OF OPERATING SYSTEMS

L:3 T:0 P:0 Credits:3

Course Outcomes: Through this course students should be able to

CO1 :: Understand the evolution of OS functionality, structures and layers

CO2 :: Explain and apply process management concepts including scheduling, synchronization, IPC, and deadlock handling.

CO3 :: Analyze deadlock conditions and apply appropriate handling techniques, and evaluate memory management strategies for efficient system performance.

CO4 :: Apply different disk management techniques

Unit I

Introduction to operating system : Introduction, Types of operating systems, System components, Operating system services, System calls

Process : Process concept, Process states, Operations on processes

Unit II

Process Management : Process control block, Context switching, Process scheduling, Interprocess communication, a case study on Windows/Linux

Unit III

CPU Scheduling : Introduction, Types of scheduling, Scheduling Criteria, Scheduling Algorithms, a case study on Windows/Linux

Process Synchronization : Background, Critical section problem, Semaphores, Concept of serializability

Unit IV

Deadlocks : Deadlock Characterization, Methods for handling deadlocks, Deadlock Prevention, Deadlock avoidance, Recovery from Deadlock, a case study on Windows/Linux

Unit V

Memory Management : logical versus physical address space, Address Binding, Dynamic Loading & Dynamic Linking, Overlays, Swapping, Contiguous Allocation, Paging, Segmentation, Segmentation with Paging, Page Replacement Algorithms, Allocation of frames, Thrashing, a case study on Windows/Linux

Unit VI

Protection : Introduction, File Access Methods, Access Matrix

Disk Management : Disk structure, disk scheduling, FCFS scheduling, SSTF scheduling, SCAN scheduling, C-SCAN scheduling, a case study on Windows/Linux

Text Books:

1. OPERATING SYSTEMS CONCEPTS by A SILBERSCHARTZ AND GALVIN, ADDISON-WESLEY

References:

1. OPERATING SYSTEMS CONCEPTS AND DESIGN by MILAN MILANKOVIC, MC GRAW HILL
2. . MODERN OPERATING SYSTEM by ANDREW S. TANENBAUM, PRENTICE HALL

